

LeetCode 1494: Parallel Courses II

Visual Notes & Concept Breakdown

1. DP State & Initialization

dp[mask] will basically give ki iss combo me phuchne ka min step kitna tha.
 tot = $1 \ll n$; (2^n combo). Har ek possible combo dekh lete.
 Agar dp[mask] == INT_MAX (yaha tak aane ka raasta ni mila abhi tk), to skip kr denge.

2. Resolving Dependencies

iss mask me jitne course le liye, unse unke dependents ki dependancy kam krenge.
 (Using indegree decrement for dependents of completed courses).

3. Identifying Available Courses

after this ho skta he kuch courses lene laayak ho gye ho.
 jiska matlab agar lene layak honge, to indegree 0 and abhi tak nhi loya hoga.

4. Optimal Greedy Choice (Available $\leq K$)

agar to less than equal to k courses lene ke laayak he, to greedily saare le lenge.

5. Combinatorial Subsets (Available $> K$)

agar k se zyada course available he to unme se k chuna padega.
 ham sirf wahi combination consider krenge jaha sirf k courses udhaye gaye.
 zyada utha nhi skte, kam me fayda nhi.

Source Code (Preserved)

```
class Solution {
public:
    int minNumberOfSemesters(int n, vector<vector<int>>& relations, int k) {

        vector<vector<int>> adj(n);
        vector<int> indegree(n, 0);

        for(auto &it : relations) {
            int u = it[0]-1;
            int v = it[1]-1;
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        adj[u].push_back(v);
        indegree[v]++;
    }

    //dp[mask] will basically give ki iss combo me phuchne ka min step kitna tha
    int tot=1<< n; //2^n combo
    vector<int> dp(tot,INT_MAX);

    dp[0]=0;

    //har ek possible combo dekh lete
    for(int mask=0; mask<tot; mask++) {

        if(dp[mask]==INT_MAX) { //yaha tak aane ka raasta ni mila abhi tk
            continue;
        }

        //iss mask me jitne course le liye
        //unse unke dependents ki dependancy kam krenge
        vector<int> currindeg=indegree;

        for(int curr=0;curr<n;curr++) {
            bool iscomp=false;
            if(mask&(1<<curr)) iscomp=true;
            if(iscomp){
                for(int it:adj[curr]) {
                    currindeg[it]--;
                }
            }
        }

        //after this ho skta he kuch courses lene laayak ho gye ho
        //jiska matlab agar lene layak honge
        //to indegree 0 and abhi tak nhi loya hoga
        vector<int> avail;
        for(int curr=0;curr<n;curr++) {
            bool took=false;
            if((mask&(1<<curr))!=0) took=true;

            if(!took && currindeg[curr]==0){
                avail.push_back(curr);
            }
        }

        //agar to less than equal to k courses lene ke laayak he
        //to greedily saare le lenge
        if(avail.size()<= k) {
            int newmask=mask;
            for(int course:avail) {
                newmask |= (1<<course);
            }
            dp[newmask] = min(dp[newmask],dp[mask]+1);
        }
    }

```

```

//agar k se zyada course available he to unme se k chuna padega
else {
    int totavail=avail.size();
    int totalCombinations = 1<<totavail; // 2^(totavail)

    for(int comb=0;comb<totalCombinations; comb++) {

        int took=0;
        for(int i=0;i<totavail;i++) {
            if((comb&(1<<i))!=0){
                took++;
            }
        }

        //ham sirf wahi combination consider kreng
        //jaha sirf k courses udhaye gaye
        //zyada utha nhi skte
        //kam me fayda nhi
        if(took==k){

            int newmask = mask;

            for(int i = 0; i < totavail; i++) {
                if((comb&(1<<i))!= 0) {
                    int tookcourse=avail[i];
                    newmask|=(1<<tookcourse);
                }
            }
            dp[newmask] = min(dp[newmask],dp[mask]+1);
        }
    }
}

return dp[tot-1];
}
};

```